# Python-Tuple

# Objective

- Concept of Tuple and Basic Operation
  - Create
  - Delete
  - Access
  - Iterate
- Basic Tuples Operator
- Built-in Tuple Functions
- Tuple Methods

# Concept of Tuple

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- A tuple is a collection which is ordered and unchangeable.
- In Python tuples are written with round brackets.
- Tuples are sequences, just like lists. The differences between tuples and lists are, the tuples cannot be changed unlike lists and tuples use parentheses, whereas lists use square brackets.
- Like string indices, tuple indices start at 0, and they can be sliced, concatenated, and so on.

```
Creating tuple:tup1 = ('physics', 'chemistry', 1997, 2000)
tup2 = (1, 2, 3, 4, 5 )
tup3 = "a", "b", "c", "d“  #you can define tuple without using () also.
tup1 = ()   #empty tuple
```

To write a tuple containing a single value you have to include a comma, even though there is only one value

tup1 = (50,);



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# Accessing Values

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## Accessing Values in Tuples

```
tup1 = ('abc', 'pqr', 1997, 2000)
tup2 = (1, 2, 3, 4, 5, 6, 7 )
print ("tup1[0]: ", tup1[0])
print ("tup2[1:5]: ", tup2[1:5])
```

**Output:**

```
tup1[0]:  abc
tup2[1:5]:  [2, 3, 4, 5]
```

## Updating Tuples

Tuples are immutable which means you cannot update or change the values of tuple elements.

# Deleting Tuple

- Unlike lists, the tuple items **cannot be deleted** by using the del keyword as tuples are immutable.
- To delete an entire tuple, we can use the del keyword with the tuple name.

```
tuple1 = (1, 2, 3, 4, 5, 6)
print(tuple1)
del tuple1
print(tuple1)
```

**Output**:

```
(1, 2, 3, 4, 5, 6)
Traceback (most recent call last):
  File "C:UsersArnavDesktopPythonsecond.py", line 4, in <module>
    print(tuple1)
NameError: name 'tuple1' is not defined
```

# Iteration

The for loop is used to iterate over the tuple elements.

```
for i in T1:
    print(i)
```

**Output:**

1
2
3
4
5



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# Basic Tuples Operator

Tuple t1 = (1, 2, 3, 4, 5) and Tuple t2 = (6, 7, 8, 9) are declared.

**Repetition**

The repetition operator enables the tuple elements to be repeated multiple times.

t1*2 = (1, 2, 3, 4, 5, 1, 2, 3, 4, 5)

**Concatenation**

It concatenates the tuple mentioned on either side of the operator.

t1+t2 = (1, 2, 3, 4, 5, 6, 7, 8, 9)

**Membership**

It returns true if a particular item exists in the tuple otherwise false

print (2 in t1) prints True.



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# Built-in Tuple Functions

1. **len(tuple)**

   It calculates the length of the tuple.

2. **max(tuple)**

   It returns the maximum element of the tuple

3. **min(tuple)**

   It returns the minimum element of the tuple.

4. **tuple(seq)**

   It converts the specified sequence to the tuple.



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# Tuple Methods

Python has two built-in methods that you can use on tuples.

1. **count()** **:-** Returns the number of times a specified value occurs in a tuple

```
thistuple = (1, 3, 7, 8, 7, 5, 4, 6, 8, 5)
x = thistuple.count(5)
print(x)
```

**Output**:
2

**2. index()** **:-** Searches the tuple for a specified value and returns the position of where it was found.

```
thistuple = (1, 3, 7, 8, 7, 5, 4, 6, 8, 5)
x = thistuple.index(6)
print(x)
```

**Output**:
4



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# Summary

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# Thank You